

USER GUIDE

ArchitectureScope

Design, audit, and document HubSpot portals — a step-by-step guide to every module.

INSIDE

- Installing and launching the app
- Connecting a HubSpot private app token
- All ten modules, step by step
- Recommended end-to-end workflows
- Saving work to the Project Library
- Troubleshooting and known limitations

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SECTION 1

What ArchitectureScope does

ArchitectureScope is a toolkit for HubSpot implementation specialists. It helps you plan a new HubSpot build, assess an existing portal, and produce the hand-off documents a project needs — all from one app that runs on your own computer.

The four areas of the app

The left-hand sidebar groups the modules by what they help you do:

Sidebar section	Modules	Use it when...
Architecture & Planning	Architecture Generator, Architecture Diagram	You're designing a new build, or want a picture of how objects connect.
Auditing	Portal Auditor, Property Audit, Executive Report	You need to assess a live portal for risk, clutter, and drift.
Documentation	Documentation Generator, Requirements Document, Joint Evaluation Plan, Test Case Document	You need records of what exists, what was agreed, and what was tested.
Library	Project Library	You want to find everything you've saved for a project.

It never changes anything in HubSpot

Read-only by design. Every HubSpot call the app makes only reads data. It never creates, updates, archives, or deletes anything in a connected portal. Even the Property Audit's 'Remove candidate' ratings are just suggestions in a spreadsheet — a person always makes the final call inside HubSpot.

Which modules need a HubSpot connection?

Module	Token needed?	Main outputs
1. Architecture Generator	No	Blueprint (.md, .json)
2. Portal Auditor	Yes	Findings (.csv, .md)
2B. Property Audit	Yes	Workbook (.xlsx), report (.pdf)
3. Documentation Generator	Yes	Documentation pack (.md, .docx)
4. Executive Report	No — reuses the pulled data	Client and internal reports (.docx)
5. Architecture Diagram	No — reuses blueprint or pulled data	.drawio, .png, .gv, .json, .xlsx, .csv

Module	Token needed?	Main outputs
6. Requirements Document	No	.docx, .xlsx, .csv, .pdf
7. Joint Evaluation Plan	No	.docx, .xlsx, .csv, .pdf
8. Test Case Document	No	.docx, .xlsx, .csv, .pdf
9. Project Library	No	Re-download anything saved

SECTION 2

Getting started

You only need to do this setup once. After that, launching the app is a single command.

Install

- 1 Make sure **Python 3.11 or newer** is installed. In a terminal, `python -V` should print 3.11 or higher.
- 2 Open a terminal in the **architecturescope** project folder.
- 3 Install the dependencies: `pip install -r requirements.txt`

Launch the app

- 1 From the project folder, run: `streamlit run Home.py`
- 2 Your browser opens automatically (usually at `http://localhost:8501`). If it doesn't, copy the address printed in the terminal into your browser.
- 3 You land on the **Home** page. Use the sidebar — or the **Start** button — to open a module.
- 4 To stop the app, return to the terminal and press **Ctrl + C**.

Tip. Try the app with zero setup first: open **Architecture Generator**, click **Load sample project**, then **Generate blueprint**. It loads a fictional gym/wellness studio project so you can see a full result immediately.

Connect to HubSpot (for auditing and documentation)

The Portal Auditor, Property Audit, and Documentation Generator read from a live portal. They authenticate with a HubSpot **private app access token**.

Step 1 — Create a private app in HubSpot

- 1 In the HubSpot portal you want to assess, go to **Settings** → **Integrations** → **Private Apps** (on newer portals this may be under **Development** → **Legacy apps**).
- 2 Create a private app and give it a clear name, e.g. "ArchitectureScope (read-only)".
- 3 On the **Scopes** tab, tick the read scopes listed in the table below.
- 4 Create the app and copy the **access token**.

Step 2 — Required scopes

What the app reads	Scopes to grant
Standard object schemas & properties	<code>crm.schemas.contacts.read</code> , <code>crm.schemas.companies.read</code> , <code>crm.schemas.deals.read</code> , <code>tickets</code>
Custom object schemas & properties	<code>crm.schemas.custom.read</code>
Record sampling (fill rates, orphan checks)	<code>crm.objects.contacts.read</code> , <code>crm.objects.companies.read</code> , <code>crm.objects.deals.read</code> , <code>crm.objects.tickets.read</code> , <code>crm.objects.custom.read</code>

What the app reads	Scopes to grant
Pipelines	Covered by the object read scopes above
Workflows	automation
Owners	crm.objects.owners.read
Teams	settings.users.teams.read

Missing a scope? The app won't crash. It shows a warning (usually an HTTP 403) naming what it couldn't pull, and builds its results from whatever it could read. HubSpot occasionally renames scopes — if a pull fails, compare with the scope picker in your portal.

Step 3 — Give the token to the app (choose one)

- **Paste it in the app (simplest).** On the Portal Auditor, Property Audit, or Documentation Generator page, paste it into **HubSpot private app access token**. It's kept only in memory for this browser session — never saved to disk or logged. You only need to paste it once per session; the other pages reuse it.
- **Use an environment file (for regular use).** Copy `.env.example` to `.env`, set `HUBSPOT_TOKEN=` to your token, and make sure it's loaded before you run Streamlit.

Keep tokens private. Never share a token or commit it to git. The `.env` file is already excluded from version control.

How data flows between modules

Modules share what you've already done in the same browser session, so you rarely repeat work:

- Pulling a portal once (from the Portal Auditor, Property Audit, or Documentation Generator) makes that **portal snapshot** available to all of them, plus the Executive Report and Architecture Diagram.
- Running the Portal Auditor's audit makes its **findings** available to the Executive Report and Architecture Diagram.
- A generated **blueprint** can be drawn in the Architecture Diagram.
- A **Requirements Document** can prefill the Test Case Document.
- Use **Refresh portal data** / **Refresh from portal** to pull fresh data. Closing or reloading the browser tab clears the session — save anything you want to keep first.

SECTION 3

Recommended workflows

Each module works on its own, but they're most useful in sequence. Pick the path that matches your project.

A. Planning a new HubSpot build

- 1 **Requirements Document** — capture goals, numbered customer requirements, terminology, risks, and open questions.
- 2 **Architecture Generator** — enter objects, pipelines, regions, integrations, volumes, and hubs; generate a blueprint.
- 3 **Architecture Diagram** — draw the blueprint, correct anything the generator got wrong, export to diagrams.net.
- 4 **Joint Evaluation Plan** — track milestones, team, and technical/business needs with the client.
- 5 **Test Case Document** — load requirements from the WRD as test cases and record QA/UAT results.
- 6 **Save to Project Library** at each step so everything lives under one project.

B. Auditing an existing portal

- 1 **Portal Auditor** — paste the token, **Pull from portal**, then **Run audit**.
- 2 **Property Audit** — **Run property audit** to rate every property; export the cleanup workbook.
- 3 **Documentation Generator** — preview and export the documentation pack (reuses the same pull).
- 4 **Architecture Diagram** — choose **Live portal snapshot** to see objects with audit findings highlighted.
- 5 **Executive Report** — download the client-ready and internal reports.

C. Handing off a finished implementation

- 1 **Documentation Generator** — produce the data dictionary, workflow inventory, association map, pipelines, and roles.
- 2 **Architecture Diagram** — export a clean .png for the hand-off deck.
- 3 **Test Case Document** — attach the final test results.
- 4 **Project Library** — collect and re-download everything for the client.

SECTION 4

Module-by-module guide

Every module below follows the same pattern: what it's for, what you need first, the steps, and what you get.

Module 1 — Architecture Generator

HUBSPOT TOKEN

Not needed

NEEDS FIRST

Nothing — start here for a new build

YOU GET

A proposed architecture blueprint (Markdown, JSON)

Turns a project's requirements into a proposed HubSpot architecture.

Steps

- 1 Enter a **Project name**. (Or click **Load sample project** to fill the whole form with an example.)
- 2 Pick **Standard objects in scope** and add any **Other custom object notes** (free text is flagged for manual review, not auto-modeled).
- 3 Choose **Regions**, **Currencies in scope**, **Integrations** (name any 'Other' integration in the notes field), and **Hubs in scope**.
- 4 Under **Estimated record volumes**, type one object per line, e.g. Contact: 45000.
- 5 Click **Generate blueprint**.

What you get

- Recommended **custom objects** and a full **data dictionary** of properties.
- An **association model** (which objects connect, with labels and cardinality).
- **Pipelines & stages**, **naming conventions**, and a base-currency recommendation.
- **Suggested workflows**, each flagged **Overwrite Risk: Yes/No** — 'Yes' means it sets a property unconditionally, so check it carefully.
- Any **warnings** appear at the top of the result — read these first.

Download as **Markdown** or **JSON**, or click **Save to Project Library**.

Adjusting the rules. The recommendations come from rules/rules.yaml. Naming conventions, pipeline templates, integration heuristics, and overwrite-risk keywords can all be tuned there without touching code.

Module 2 — Portal Auditor

HUBSPOT TOKEN

Required

NEEDS FIRST

A private app token

YOU GET

A filterable list of best-practice findings (CSV, Markdown)

Connects to a live portal and checks it against best practice.

Steps

- 1 Optionally enter a **Project name** (needed only to save to the library).
- 2 Paste the **HubSpot private app access token** (skip if it's already in the session or in .env).
- 3 Click **Pull from portal** and wait while schemas, properties, pipelines, workflows, owners, and teams load.
- 4 Click **Run audit**.
- 5 Narrow the list with **Filter by severity** and **Filter by area**.
- 6 Download **CSV** or **Markdown**, or **Save to Project Library**.

What it checks

Check	What it flags
Duplicate properties	Same label with different internal names; near-duplicate picklist options
Unused properties	No value in any sampled record
Missing required fields	Baseline-required properties empty on sampled records
Orphaned records	Custom-object records with no association (sampled)
Risky workflows	Re-enrollment on + sets a property with no visible condition
Permission anomalies	Owners with no team; teams with no members
Naming drift	Names that break the same conventions Module 1 uses

About sampling. Record-level checks look at a sample (100 records per object by default) and say so in each finding. Treat them as strong signals, not a full census.

Module 2B — Property Audit

HUBSPOT TOKEN

Required (or reuse a pull)

NEEDS FIRST

A portal snapshot — pull here or on any other portal page

YOU GET

Keep / Review / Remove rating for every property; Excel workbook + PDF report

Rates **every property on every object** so you can plan a property cleanup.

Steps

- 1 Paste the token and click **Pull from portal** (skip if you already pulled on another page).
- 2 Click **Run property audit**.
- 3 Review **Summary by object** (worst first), then **Flagged for action** — your cleanup worklist. Use **Filter by object** to focus.

- 4 Download the **.xlsx** workbook and **.pdf** report, or **Save to Project Library**.

How ratings are decided

Two signals are measured for each **custom** property: **Fill %** (share of sampled records with a value) and **Uses** (number of workflows that reference it). HubSpot's own native properties are always **Keep**.

Rating	Rule
Keep	Native property, or custom with 25%+ fill and at least one workflow reference.
Review	Custom, and either used by a workflow but under 25% filled, or 5%+ filled but used by no workflow.
Remove candidate	Custom, under 5% filled, and referenced by no workflow.

Using the workbook

- **Summary** — counts and a cleanup score per object.
- **Flagged for Action** — only the Review/Remove rows. Work through this list.
- **All Properties** — every property, with blank **Notes** and **Tag** columns for your review.
- **How to use** — a reminder of the rules for anyone you share it with.

Verify before deleting. 'Uses' only counts workflow references — the app doesn't read forms, lists, reports, or dashboards. A 'Remove candidate' may still be used in one of those. Treat the list as a starting point to check, never an automatic delete list.

Module 3 — Documentation Generator

HUBSPOT TOKEN

Required (or reuse a pull)

NEEDS FIRST

A portal snapshot

YOU GET

Documentation pack (Markdown, styled .docx)

Steps

- 1 Paste the token (if needed) and click **Pull from portal**.
- 2 Preview each section on the page — expand an object in the **Data dictionary** to see its properties.
- 3 Download **Markdown** or **.docx**, or **Save to Project Library**.

What's in the pack

- **Data dictionary** — every object with its properties, types, and options.
- **Workflow inventory** — every workflow found.
- **Association map** — connections declared on custom objects.
- **Pipelines & stages** and a **Roles & permissions** summary (teams and owners).

Module 4 — Executive Report

HUBSPOT TOKEN

Not needed

NEEDS FIRST

A portal pull; ideally also a Portal Auditor audit

YOU GET

Client report and internal report (.docx)

Turns the portal data and audit findings into a plain-language report.

Steps

- 1 Pull the portal (Module 2 or 3) and, ideally, **Run audit** in the Portal Auditor.
- 2 Open **Executive Report** and optionally type a **Client / project name** to personalize the title.
- 3 Review the on-screen preview: overall health rating, what's in the portal, pipelines, automation & people, and things worth fixing.
- 4 Download the report that fits your audience.

Download	Audience	Contents
Client report (.docx)	The client	Polished, plain language. Up to 3 examples per finding category ('...and N more like this').
Internal report (.docx)	Your team	Every finding in tables, property-group breakdowns, association IDs, area × severity summary.

The **health rating** (Excellent / Good / Fair / Needs Attention) counts only High and Medium findings — Low items are cosmetic. If no audit has been run, the report says so rather than claiming a clean bill of health.

Module 5 — Architecture Diagram

HUBSPOT TOKEN

Not needed

NEEDS FIRST

A blueprint (Module 1) and/or a portal pull

YOU GET

Editable diagram: .drawio, .png, .gv, .json, .xlsx, .csv

Steps

- 1 Choose the **Diagram source: Live portal snapshot** (the real portal) or **Architecture blueprint** (the proposed design).
- 2 For a live portal, leave **Include HubSpot activity objects** off unless you need Tasks, Notes, Emails, etc. — they add clutter.
- 3 Fix anything wrong in the **Objects** and **Connections** tables: rename, change kind or cardinality, add or delete rows, mute a finding. The diagram updates live.
- 4 Use **Focus diagram on** to show only selected objects, and untick **Show connection labels** if it's busy.
- 5 Enter a **Project name** and export, or **Save to Project Library**.
 - Objects with audit findings get a colored border: **red** = High, **orange** = Medium, **yellow** = Low.
 - **Reset to generated data** discards your table edits and rebuilds from the source.

- The **.drawio** file opens free at app.diagrams.net for drag-and-drop editing; the **.png** is ready for slides.
- Saved diagrams can be reopened later from **View a previously saved diagram** — no re-pull needed.

Modules 6–8 — Project documents

Three note-taking modules structured like HubSpot's own templates. They need no token and no other module. Every table is editable: click a cell to type, use the **+** row at the bottom to add a row, and select rows to delete them. Each exports to **.docx**, **.xlsx**, **.csv**, and **.pdf**.

Module 6 — Requirements Document (WRD)

- **Project Overview** — purpose and goals.
- **Customer Requirements** — numbered; the order matters because the Test Case Document can load them.
- **Project Definitions** — the client's own terminology.
- **Known Challenges or Risks**, **Project Plan** (a draft, not a commitment), **Open Questions**, **Appendix and Resources**.

Module 7 — Joint Evaluation Plan

- Prospect's company name and solutions-partner contact details.
- **Milestones** (target date, status, owner, outcome), **Team Members**, **Technical Needs**, **Business Needs**, **Questions**.

Module 8 — Test Case Document

- 1 Enter a **Project name**.
- 2 If you've filled in a Requirements Document this session, click **Load requirements from Requirements Document** to start with one row per requirement.
- 3 For each test case, fill in steps, expected result, actual result, test round, and feedback.
- 4 Export or **Save to Project Library**.

Note. Loading requirements is a one-time copy. Later edits to the WRD won't change rows already loaded into the Test Case Document.

Module 9 — Project Library

Every module has a **Save to Project Library** button beside its downloads. It becomes active once a project name is entered, and saves every export format at once, stamped with the date and time.

- 1 Open **Project Library** and pick a **Project**.
- 2 If a diagram was saved, a **Visual preview** shows it exactly as saved.
- 3 Browse files grouped by document type, then **Download** or **Delete** any file.

Files are stored on your computer under `project_library/<project>/<document type>/`. The exact location is shown at the top of the page and can be changed with the `PROJECT_LIBRARY_DIR` environment variable.

Tip. Use exactly the same project name in every module so all documents land in one project folder.

SECTION 5

Troubleshooting & FAQ

Problem	What to do
streamlit is not recognized	Dependencies aren't installed in the active Python. Run <code>pip install -r requirements.txt</code> again, or use <code>python -m streamlit run Home.py</code> .
HTTP 401 when pulling	The token is wrong or revoked. Copy a fresh token from the private app.
HTTP 403 / 'scope likely missing'	Add the missing read scope to the private app (Section 2), then click Refresh portal data .
Pull is slow	Large portals take time; the app automatically waits and retries when HubSpot rate-limits it.
Executive Report or Diagram says 'nothing yet'	Pull a portal (or generate a blueprint) first in the same browser session.
Findings missing from the report or diagram	Run the audit in the Portal Auditor ; the Property Audit ratings are separate.
Save to Project Library is greyed out	Enter a project name on that page first.
My work disappeared	Reloading or closing the tab clears the session. Save to the Project Library or download before leaving.
Newer workflows aren't listed	Only workflows visible through HubSpot's legacy Automation API are read; custom-coded/newer flows may be missing.

Known limitations

- Workflow inspection uses HubSpot's legacy v3 Automation API — newer 'flows' and custom-coded actions aren't parsed.
- The association map only shows associations declared on custom-object schemas, not default standard-to-standard ones.
- The orphan check follows only each custom object's first declared association.
- Record-level checks are sampled (100 records per object by default; adjustable in `rules.yaml`).
- Property Audit 'Uses' counts workflow references only — not forms, lists, reports, or dashboards.
- Blueprint recommendations are simple rule-based pattern matches — always review them with your own judgment.

Quick reference: button names

Button	Where	What it does
Load sample project	Architecture Generator	Fills the form with a demo project
Generate blueprint	Architecture Generator	Builds the proposed architecture
Pull from portal / Refresh	Auditor, Property Audit, Doc Generator	Reads the portal (read-only)
Run audit	Portal Auditor	Runs the best-practice checks
Run property audit	Property Audit	Rates every property
Reset to generated data	Architecture Diagram	Discards table edits
Load requirements from Requirements Document	Test Case Document	Prefills test cases
Save to Project Library	Every module	Saves all formats to disk